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RELIABILITY STUDY

Reproducibility of the pressure biofeedback unit in measuring transversus abdominis muscle activity in patients with chronic nonspecific low back pain

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Summary The Pressure Biofeedback Unit (PBU) is often used by clinicians and researchers to indirectly evaluate transversus abdominis (TrA) muscle activity. The purpose of this study was to evaluate the inter and intra-examiner reproducibility of the PBU in measuring TrA muscle activity in fifty patients with chronic nonspecific low back pain. This study was performed using a test-retest design with a seven day interval. An Intraclass Correlation Coefficient ($ICC_{2,1}$) of 0.74 (95% CI 0.54 to 0.85) and 0.76 (95% CI 0.58 to 0.86) was observed for the intra and inter-examiner reproducibility, respectively. The intra-examiner agreement (Limits of Agreement - LOA = 2.1 to −1.8 mmHg) and the inter-examiner agreement (LOA = 2.0 to −1.9 mmHg) were within the limits of agreement on 95% of occasions. The reproducibility of PBU in measuring TrA muscle activity in patients with chronic nonspecific low back pain ranged from satisfactory to excellent.

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Introduction

The use of motor control exercises has become very popular in clinical practice for the treatment of nonspecific low back pain (Ferreira et al., 2007, 2006; Macedo et al., 2008, 2009). The rationale for the use of motor control exercises is that the deep muscles of the abdominal wall have a key role in the dynamic control of the lumbar spine (Hodges and Richardson, 1996, 1998). For example, a delay in the onset of activation of the transversus abdominis muscle (TrA) has been observed in patients with recurrent low back pain compared to asymptomatic controls (Hodges and Richardson, 1998, 1999).

Most of the studies that have measured the activity of the deep abdominal wall muscles used fine-wire electromyography (Beith et al., 2001; Hodges and Richardson, 1996). However, this type of assessment is invasive, painful, uncomfortable, expensive and may present the risk of infection (Costa et al., 2004, 2009). Thus, the Pressure Biofeedback Unit (PBU) has been developed as an alternate approach to indirectly measure TrA muscle activity (Chattanooga, 2005; Costa et al., 2004, Hodges et al., 2003). In rehabilitation the PBU is usually used to evaluate the activity of the abdominal and neck muscles as well as provide biofeedback to patients undertaking motor control exercise interventions (Cynn et al., 2006).

The PBU (Figure 1) is a simple pressure transducer consisting of a three-chamber air-filled pressure bag, a catheter and a sphygmomanometer gauge (Chattanooga, 2005). The pressure bag is 16.7 $\times$ 24 cm in size and made from inelastic material. The sphygmomanometer has a range of 0 mmHg–200 mmHg, with 2-mmHg intervals on the scale.

Figure 1 Pressure Biofeedback Unit (STABILIZER®).

In order to be useful, PBU measures need to have acceptable clinimetric properties such as reproducibility. Reproducibility is the degree to which repeated measurements provide similar results in clinically stable patients and serves as an umbrella term for reliability and agreement (Terwee et al., 2007). Agreement can be defined as the degree to which scores on repeated measurements are close to each other (i.e. absolute measurement error) while reliability can be defined as the degree to which individuals can be distinguished from each other, despite the measurement error (i.e. relative measurement error) (de Vet et al., 2006; Terwee et al., 2007).

The clinimetric properties of measuring PBU in the TrA muscle activity were extensively discussed in a recent published systematic review (Lima et al., 2011). Previous studies (Costa et al., 2004, 2006; Figueiredo et al., 2005; Hodges and Richardson, 1996; Storheim et al., 2002, von Garnier et al., 2009) that evaluated the reproducibility of PBU in measuring TrA muscle activity answered some, but not all questions about the use of this test to guide the clinical management of low back pain; a limitation is that most of them solely sampled healthy individuals (Costa et al., 2004, 2006; Figueiredo et al., 2005; Storheim et al., 2002). Only the studies conducted by Hodges et al. (1996) and von Garnier et al. (2009) recruited volunteers with a previous history of low back pain. Although some authors have concluded that PBU provides reproducible measures of TrA muscle activity, their results are questionable due to suboptimal research designs, such as small sample sizes; lack of standardization of the assessment protocol; recruitment of healthy subjects and insufficient statistical power (Costa et al., 2004, 2006; Figueiredo et al., 2005; Storheim et al., 2002, von Garnier et al., 2009). While the reproducibility of PBU measures has already been evaluated in previous studies, the lack of consensus amongst studies means that the current evidence is not conclusive. Moreover, just a few studies have sampled participants with low back pain, which is one of the target conditions for the use of PBU.

Therefore, this study aimed to evaluate the intra and inter-examiner reproducibility of PBU in measuring TrA muscle activity in patients with chronic nonspecific low back pain.

Methods

This study used a test-retest design (Hulley et al., 2008), and was carried out at the Laboratory of Kinesiology and Functional Assessment of the Department of Physiotherapy of the Federal University of Pernambuco, Brazil in 2009.

The sample was selected from students of the Federal University of Pernambuco who presented with symptoms of chronic nonspecific low back pain. The required sample size was estimated based upon the recommendations of specific guidelines on clinimetric properties that suggest a sample of at least 50 participants for reproducibility studies (Terwee et al., 2007).

We included participants with an episode of chronic nonspecific low back pain (*i.e.*, low back pain of duration of at least three months). We excluded participants who had acute low back pain (*i.e.*, an episode of low back pain of less than 6 weeks), those who were pregnant, who had previous abdominal wall or spinal surgery, with a body mass index (BMI) of 25 or over, were menstruating during the tests and those with suspected or confirmed serious pathologies (*i.e*, nerve root compromise; fractures; cancer; infectious diseases of the spine; cauda equina syndrome and widespread neurological disorders). Participants who missed consecutive tests were also excluded.

Measurements were performed by two physiotherapists in order to test the inter-examiner reproducibility of PBU. The test-retest interval was approximately 5 min and the order of examiners was randomized. In order to avoid exchanging of information between the examiners, none of them was present during the measures performed by the other examiner. For the study of intra-examiner reproducibility, one of the examiners conducted two measurements of the TrA muscle activity on two different occasions, with an interval of seven days to reduce recall. Examiners A and B were physiotherapists with extensive experience using the PBU. The correct use of PBU was important (Cairns et al., 2000; Richardson and Jull, 1995) therefore both examiners received additional training for the use of the PBU in order to optimize the standardization of the tests. They were also blinded to the measures collected from each other (Terwee et al., 2007).

All participants received basic information about the anatomy, biomechanics and function of the TrA muscle, as well as about the procedure of testing and training the TrA muscle contraction. All subjects were previously instructed to fast for 2 h prior to testing (including water), empty the bladder immediately before the tests and not perform abdominal exercises prior to the tests (Costa et al., 2004). For both tests, participants and examiners have adopted the same clinical, environmental and temporal conditions to avoid external influences or internal errors during the period of data collection. Participants were positioned in a prone position on a hard surface, with the lower limbs positioned with the feet off the plinth and with the arms beside the trunk.

Thereafter, the inflatable bag from the PBU was placed between the anterior superior iliac spine and the navel. Before starting the contractions, the bag was inflated to a pressure of 70 mmHg with the valve closed. Participants were instructed to breathe using mainly the abdominal wall and then the inflatable bag was adjusted to 70 mmHg again. Patients were requested to perform three TrA muscle contractions with the following verbal commands standardized by the examiner: "Draw in your abdomen without moving the spine or pelvis" and maintain these contractions for 10 s (Costa et al., 2004). According to the manufacturer of the PBU (Stabilizer®, Chattanooga Group Inc., Hixson, TN, USA), the ability to contract the TrA muscle results in a pressure reduction from 4 to 10 mmHg, which is recorded by the pressure gauge of PBU. Thus, based on this information and previous research on the topic, a pressure reduction of at least 4 mmHg was defined as a successful result (Costa et al., 2004, Hodges et al., 1996).

All information was recorded using a digital evaluation form provided by a specific software (Miograph®). Pain intensity was measured by the Pain Numerical Rating Scale and disability was measured by the Roland Morris Disability Questionnaire, both adapted and clinimetrically tested for Brazilian-Portuguese speakers (Costa et al., 2007, 2008). These outcome measures were administered on the first day, prior to the PBU test.

The sample characteristics are present using descriptive statistics. To describe the reliability, we used the Intraclass Correlation Coefficient ($ICC_{2,1}$); ICC values lower than 0.4 can be classified as poor, between 0.4 and 0.7 can be classified as satisfactory and over 0.7 can be classified as excellent (de Vet et al., 2006). We used three measures of agreement: the Bland and Altman plots (Bland and Altman, 1986), the Standard Error of the Measurement (SEM) and the Smallest Detectable Change (SDC). The SEM was calculated by dividing the standard deviation of the mean differences between the two measurements by the square root of 2 (SD differences/$\sqrt{2}$) and the SDC was calculated using the formula $SDC = 1.96 \times \sqrt{2} \times SEM$. The SEM reflects the absolute error of the instrument and the SDC reflects the smallest within person change in a score that can be interpreted as a ''real'' change, above the measurement error one of an individual (de Vet et al., 2006). Data were analyzed with SPSS software (Statistical Package for Social Sciences) version 15.0.

This study was approved by the Institutional Research Ethics Committee (#00490236000-09).

Results

In total, 57 participants were tested, of whom seven were excluded for not attending the retest, therefore 50 individuals were analyzed. The sample characteristics are presented in Table 1. Of the 50 participants, 38 (76%) were women and were on average 22 years old, weighed 63.7 kilos and had an average height of 1.70 m. The mean low back pain duration was 1.9 years, the mean low back pain intensity measured by the Pain Intensity Numerical Rating Scale was 5.1 points and the mean disability measured by the Roland Morris Disability Questionnaire was 9.1 points.

Examiner A obtained a mean pressure value of −4.1 mmHg (SD 2.4) and −4.9 mmHg (SD 2.9) on the first (Test 1) and second test (Test 2), respectively. Examiner B obtained a mean of −4.7 mmHg (SD 3.1). The mean difference of the Examiner A (Test 1 - Test 2) was 0.8 mmHg (SD 2.3) reaching an $ICC_{2,1}$ of 0.74 (95% CI 0.54 to 0.85). The mean difference between examiners A and B was 0.2 mmHg

Table 1 Sample characteristics.

Variables		
Gender (N%)	Female	38 (76%)
	Male	12 (24%)
Age (years)		22 SD 2.3
Low back pain duration (years)		1.9 SD 3.4
Weight (kg)		62.7 SD 12
Height (m)		1.7 SD 0.1
Pain Intensity Numerical Rating Scale (0–10)		5.1 SD 1.8
Roland Morris Disability Questionnaire (0–24)		9.1 SD 5.9

Table 2 Reliability and agreement estimates for pressure measures.

Measures	Mean (mmHg)	Minimum/Maximum (mmHg)	$ICC_{2,1}$ (95% CI)	SEM (mmHg)	SDC (mmHg)
PBU Examiner A (Test 1)	−4.1SD 2.4	−10/3	–		
PBU Examiner A (Test 2)	−4.9SD2.9	−13/2	–		
PBU Examiner B	−4.7SD 3.1	−12/3	–		
Examiner A (Test 1 – Test 2)	0.8 SD 2.3	0.1/1.4	0.74 (0.54–0.85)	1.62	4.49
Examiner A – Examiner B	0.2 SD 2.6	−0.8/0.6	0.76 (0.58–0.86)		

*ICC = Intraclass Correlation Coefficients; SEM = Standard Error of the Measurement; SDC = Smallest Detectable Change.

(SD 2.6) reaching an $ICC_{2,1}$ of 0.76 (95% CI 0.58 to 0.86). The SEM was 1.62 mmHg and SDC = 4.49 mmHg (Table 2).

The limits of agreement for the intra-examiner agreement of Examiner A ranged from 2.1 to −1.8 mmHg while the limits of agreement for the inter-examiner agreement ranged from 2.0 to −1.9 mmHg, as shown by the Bland and Altman plots (Figure 2 and 3).

Discussion

Our study shows that measurements of TrA muscle activity using the PBU, during a voluntary contraction maneuver in patients with chronic nonspecific low back pain, showed reproducibility estimates that ranged from satisfactory to excellent for both intra and inter-examiner conditions.

The use of the PBU by physiotherapists as a feedback tool for the evaluation of chronic nonspecific low back pain patients has increased over the last decade. In clinical practice, it is common for patients to be evaluated several times by the same or by different examiners. Therefore, it is important to know the reproducibility of measures and instruments used by the same examiner on different occasions as well as by different examiners (Portney and Watkins, 2008).

With regards to intra-examiner reliability, the $ICC_{2,1}$ value of 0.74 represents excellent reliability for the measurements of the TrA muscle and similar results were found in most of the studies on this topic (Costa et al., 2004, 2006; Figueiredo et al., 2005, von Garnier et al., 2009). However, this finding was not consistent, given the results obtained by Storheim et al. (2002) in which the intra-examiner reliability was considered poor. This discrepancy may be due to methodological differences between studies, such as sample sizes, study participants, standardization of breathing during the tests and different methods of statistical analysis. Storheim et al. (2002) recruited a sample of 15 healthy subjects, did not standardize the breathing during the tests and the data analysis was performed using the coefficient of variation, unlike the present study that used the Intraclass Correlation Coefficient. Likewise, the inter-examiner reliability of the PBU achieved excellent results ($ICC_{2,1} = 0.76$), corroborating with findings obtained by Figueiredo et al. (2005) who found an ICC of 0.82. However, our results were different than the ones obtained by von Garnier et al. (2009) with ICCs ranging from 0.20 to 0.68 (poor to satisfactory).

Although some caution was taken in order to reduce the error of measurements of the PBU (extensive training of examiners and the use of the same equipment in all measurements), some conflicting results were observed when compared with some previously published studies. Perhaps these differences with regards to the reliability of PBU were related to the different criteria adopted in each study to define how the pressure data was collected. While Richardson et al. (2004) collected a pressure reduction of 4–10 mmHg for 10 s, Costa et al. (2006) and Storheim et al. (2002) recorded the maximum pressure reduction of at least 2 s within a period of 8–10 s. In contrast, von Garnier et al. (2009) performed their data collection using a set of four criteria that participants would have to fulfill for the correct TrA muscle contraction: continuous breathing, absence of muscle substitution maneuvers, appropriate muscle contraction checked by palpation test and a pressure reduction of at least 1 mmHg for 4 s within a period of 10 s. In summary, it can be observed that some authors collected only the peak of muscle contraction while others have collected the most stable value for a certain period time. The results of this study showed that approximately 40% of participants were not able to achieve a successful contraction which is

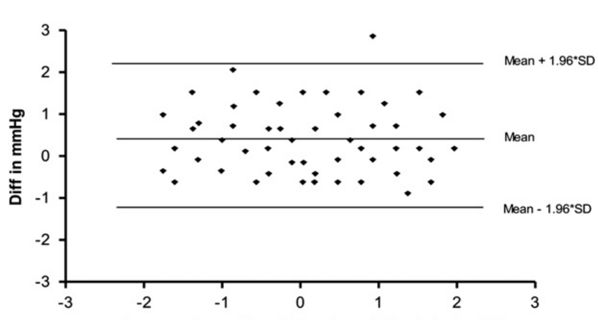

Figure 2 Bland and Altman plots for pressure measures of Examiner A (test 1 – test 2).

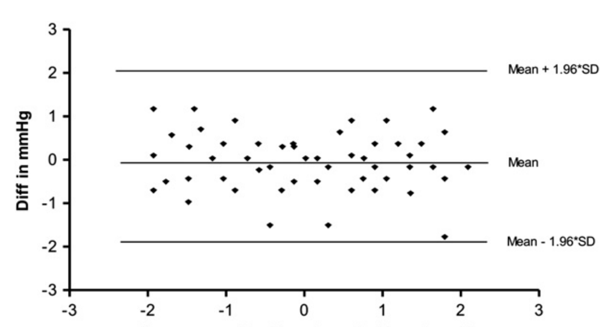

Figure 3 Bland and Altman plots for pressure measures of Examiner A and Examiner B.

similar to previous studies where it was found that people with history of chronic nonspecific low back pain have difficulty to perform a correct contraction of the TrA muscle (Gouveia and Gouveia, 2008; Hodges, 1999; Richardson and Jull, 1995). Cairns et al. (2000) evaluating two groups (symptomatic and healthy subjects) with the PBU observed that the pressure results obtained for patients with low back pain were different from those obtained for the healthy subjects, confirming the importance of testing the clinimetric properties of the PBU in a homogeneous sample composed of patients with chronic nonspecific low back pain.

The adoption of different criteria for pressure data collection in some studies might cause a tendency to underestimate the reliability of the PBU. Considering the information provided by manufacturer of the PBU, the literature and different criteria used in previous studies, the most appropriate criteria to determine success of this test remain unclear. Moreover, the PBU manufacturer claims an accuracy rate of plus/minus 3 mmHg and we have established as a successful result a pressure reduction of 4 mmHg (Chattanooga, 2005). Another fact to consider is the standardization in the assessment protocol, which has shown to improve the reproducibility of other processes of evaluation in patients with low back pain (Chiradejnant et al., 2003; Maher and Adams, 1994; Maher et al., 1998).

Most of the studies about the reproducibility of the PBU used 7 dayas the between-test interval (Costa et al., 2004, 2006; Figueiredo et al., 2005; Storheim et al., 2002) which was the same interval used in our study. Time periods between the repeated measurements should be long enough to avoid recall of data by the investigators, but short enough to ensure there are no clinical changes in the participants (Terwee et al., 2007). However, in all analyses of assessment instruments it is possible that learning effects could influence the results on the second day of tests and as a consequence has the potential of influencing the reproducibility estimates (Thomas et al., 2005).

In order to facilitate the comparison of results of this study with the majority of investigations about clinimetric properties, we decided to estimate the reliability from the average of three measures. This fact could constitute a potential difficulty in generalizing our findings because, in clinical practice, it is usual to consider only a single PBU measure. However there is evidence that use of repeated measures is the best way to reduce error and consequently increase the reproducibility (Chiradejnant et al., 2003; Costa et al., 2009).

The findings of Bland and Altman plots showed excellent intra-examiner agreement (Limits of Agreement - LOA = 2.1 to −1.8 mmHg), indicating that measures related to the first test were in agreement with the second test in 95% of occasions. Similarly, we found excellent inter-examiner agreement (LOA = 2.0 to −1.9 mmHg), which means that measures relating to Examiner A were in agreement with the Examiner B in 95% of occasions. Our results also showed a SEM of 1.62 mmHg and an SDC of 4.49 mmHg, which means that the absolute measurement error of the PBU is around 1.62 mmHg, and there would need to be at least a 4.49 mmHg improvement in the TrA muscle activity to make sure that a true change has occurred (Costa et al., 2009, de Vet et al., 2006). Although considering agreement values (SEM and SDC) for pressure measurements, it is important to consider that there is no normative data in the literature that could allow the determination of the minimal important change (MIC). Ideally would be that the SDC is smaller than MIC (de Vet et al., 2006), however these estimates are not yet available in the literature.

The influence of breathing on the evaluation of patients with chronic nonspecific low back pain with PBU was unknown until recently. Lafond et al. (2009) investigated this influence for the clinical assessment of lumbar postural control and concluded that there are significant differences between pressure measurements collected during breathing and apnea, with higher values observed during normal breathing. They also found that at least three repeated measures were needed in order to ensure acceptable reliability during apnea. There is a tendency that participants without guidance with regards to normal breathing, to contract the TrA muscle associated with apnea (Cairns et al., 2000). However, it has been demonstrated that data collection performed at the end of expiratory time could reduce contamination of results (Hodges and Richardson, 1997).

It has been suggested that even with extensive training it would be difficult to contract a single muscle (Beith et al., 2001). There is apparently no support from research that TrA muscle can be isolated and activated on its own and it is likely that the novice patient is more likely to contract wide groups of abdominal muscles. This tends to decrease the utility of measuring TrA muscle in isolation and this can be considered as a limitation of the study (Cholewicki et al., 2002; Sapsford et al., 2001; Urquhart et al., 2005a,b). Furthermore while there have been reports of asymmetry of TrA muscle activity in low back pain patients (Allison et al., 2008; Bjerkefors et al., 2010), however we are unable to contribute to this area of research because we evaluated TrA muscle activity bilaterally.

Conclusion

The intra and inter-examiner reproducibility of measures of PBU in the TrA muscle activity, during voluntary contraction maneuver in patients with chronic nonspecific low back pain ranged from satisfactory to excellent. It is suggested that other clinimetric properties (such as validity) of PBU should be an important topic for future investigations.

Conflict of interest

There are no conflicts of interests.

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